SCIENCE · SYSTEMS AT WORK

One Team, Many Parts

How plant organs move materials together, and what happens to the body when one part is damaged

Plant transport | Torn ACL | Asthma attack



WARM-UP

Three mysteries



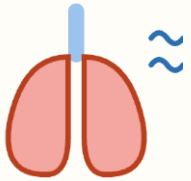
The droopy pechay

At noon on a hot day, the pechay in the school garden droops. By evening, it stands up again.



The pop in the knee

During a game, Paolo lands, twists, and hears a “pop.” Then his knee gives way.



The tight chest

During PE, Bea starts wheezing. Every breath feels like sucking air through a tiny straw.

What do these three stories have in common?

LEARNING GOALS

Your mission today

1. Explain how **roots, stems, and leaves work together** to move water, minerals, and food through a plant.

2. Describe how a **torn ACL** affects the skeletal and muscular systems, and the whole body.

3. Describe how an **asthma attack** affects the respiratory and circulatory systems, and the whole body.

Key words: root hairs · xylem · phloem · transpiration · ligament · tendon · ACL · bronchioles · alveoli · inhaler

THE BIG IDEA

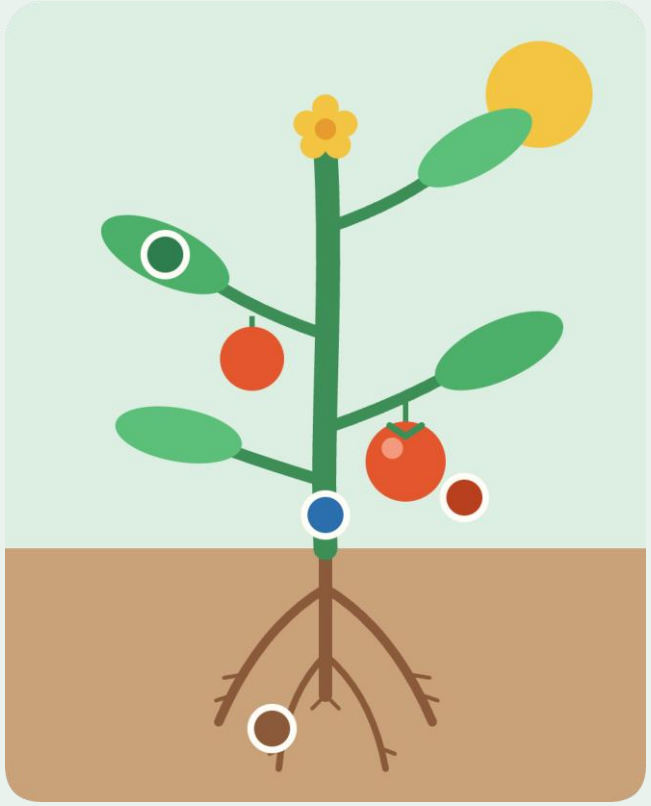
Teams within teams

Level	In a plant	In a human
Cells	Root hair cell	Muscle cell
Tissues	Xylem tissue	Muscle tissue
Organs	Root, stem, leaf	Lungs, heart, bones
Organ systems	Root system, shoot system	Skeletal, muscular, respiratory, circulatory
Organism	The whole tomato plant	The whole person

Think of a basketball team: players form positions, and positions form a team. If one player gets hurt, the whole team plays differently.

PART 1 · PLANT TRANSPORT

Meet Kamatis and its team



Story: Kamatis is the tomato plant in our school garden. Its organs keep it alive, and none of them can work alone.

Roots

Anchor the plant and absorb water and minerals from the soil.

Stem

Holds the plant up and carries materials between roots and leaves.

Leaves

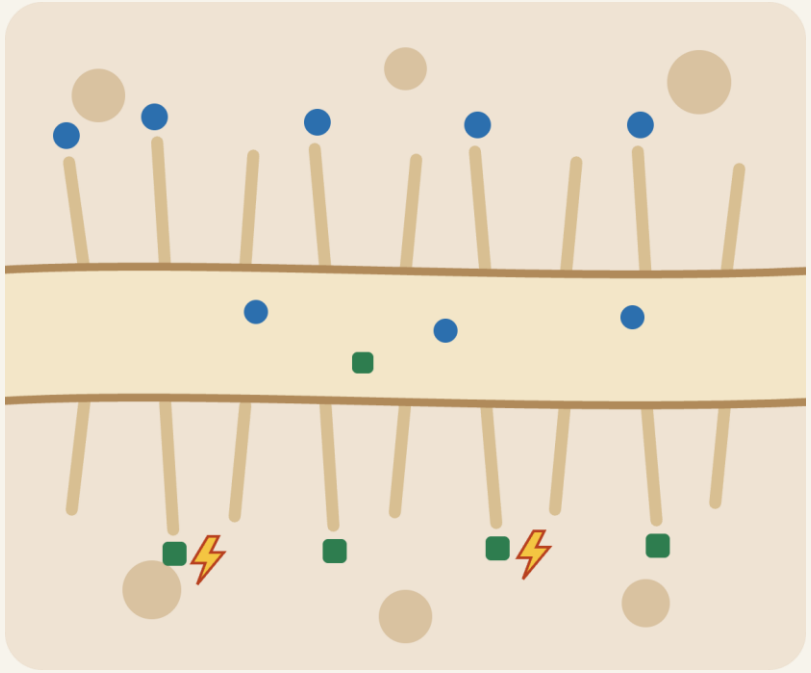
Make food (sugar) by photosynthesis and exchange gases with the air.

Flowers and fruits

Make seeds and store sugar. That’s why ripe tomatoes taste sweet!

PART 1 · ROOTS

Roots: the water collectors



Root hairs up close · **blue = water** · **green = minerals**

Root hairs are tiny outgrowths of root cells. Millions of them give a huge surface area.

Water moves from the wet soil into the root hairs by **osmosis**. No energy needed.

Minerals like nitrogen and potassium are pumped in by **active transport**, which uses ATP.

Think of a mop

Thousands of thin strands soak up far more water than one thick stick could.

PART 1 · STEM

The stem’s two-way highway

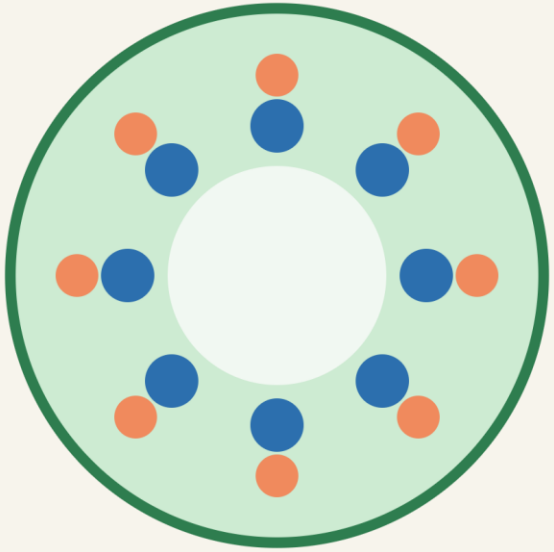
Xylem

Carries: water and minerals

Direction: up only, from roots to leaves

Made of: hollow, non-living tubes

Like the water pipes in your house



Stem cross-section · **blue = xylem** · **orange = phloem**

Phloem

Carries: sugar (food)

Direction: up and down, wherever food is needed

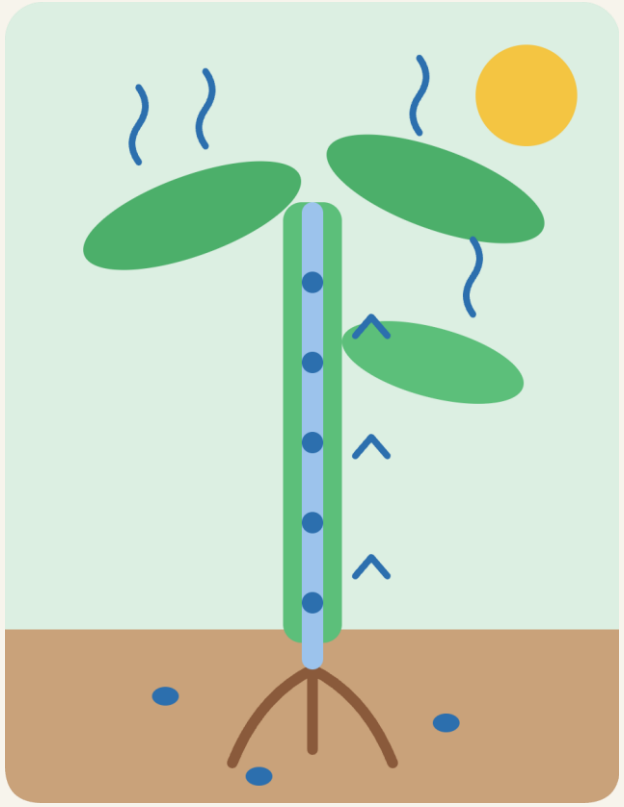
Made of: living tube cells

Like delivery riders bringing food to every barangay

Memory trick: **ph**loem carries **f**ood. Both start with an “f” sound!

PART 1 · MOVING WATER UP

How does water climb so high?



1. **Leaves pull.** Water evaporates out through tiny pores called stomata. This is **transpiration**.

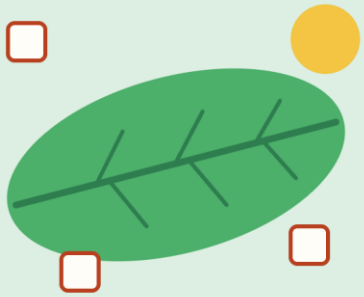
2. **Water sticks together.** Like a chain of hands, the water in the xylem rises as one long column.

3. **Roots refill.** More water enters the roots to replace the water the leaves lost.

Like sipping juice through a straw: your mouth is the leaves, the straw is the xylem, and the juice is the water in the soil.

PART 1 · MOVING FOOD

Sugar on the move: source to sink



Source: the leaves

Leaves make sugar by photosynthesis and load it into the phloem.

Sink: fruits

Sugar makes ripe tomatoes and mangoes sweet.

Sink: roots

Roots use sugar for energy and store extra as starch, like in camote and cassava.

Sink: growing tips

New leaves, buds, and flowers use sugar as energy to grow.

Phloem carries sugar **up or down**: from where it is made (the **source**) to where it is used or stored (the **sink**).

PART 1 · TEAMWORK

No organ works alone

Water and minerals travel UP through the xylem



Leaves need water from the roots to make sugar. They send sugar down.

The stem holds both pipes and lifts the leaves up toward the light.

Roots need sugar from the leaves for energy. They send water up.

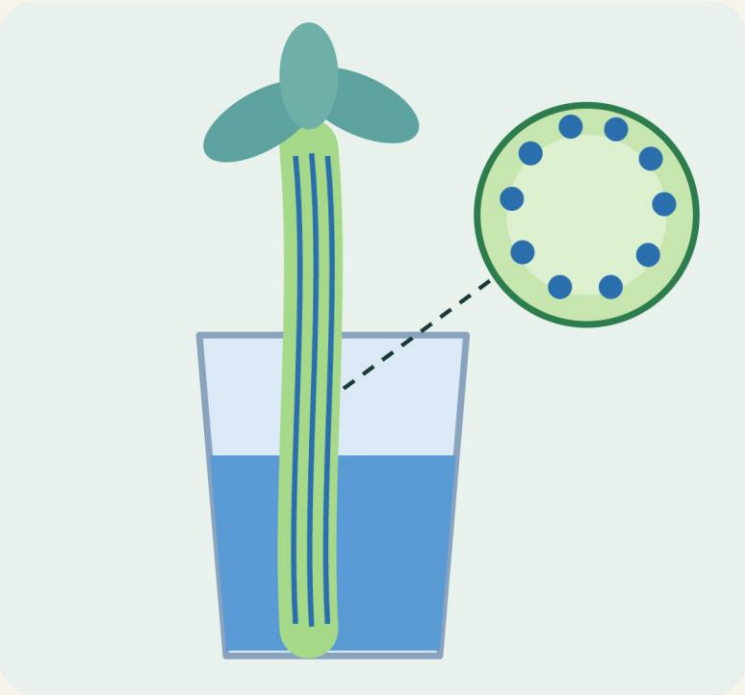


Sugar travels DOWN to the roots through the phloem

If one organ fails, the others soon struggle too.

PART 1 · TRY IT!

Experiment: the blue celery



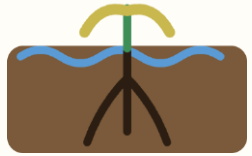
Blue dots in the cut stalk = **xylem tubes**

1. Fill a clear glass halfway with water. Add about 10 drops of blue food coloring.
2. Trim the bottom of a fresh celery stalk with leaves, or a white flower such as a carnation.
3. Stand it in the colored water. Wait 4 to 24 hours.
4. Cut across the stalk. The colored dots you see are the xylem tubes!

Predict first: What color will the leaves be tomorrow? Why?

PART 1 · WHEN SOMETHING GOES WRONG

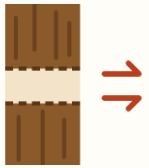
When one plant part is damaged



Rotten roots

Kamatis was overwatered, and its roots began to rot.

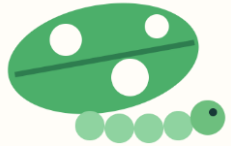
Effect: Leaves wilt, even in wet soil. Rotten roots can’t absorb water.



A ring of missing bark

A goat chewed a ring of bark all around a young tree’s trunk.

Effect: The phloem in the bark is cut. The roots starve, and the tree slowly dies.



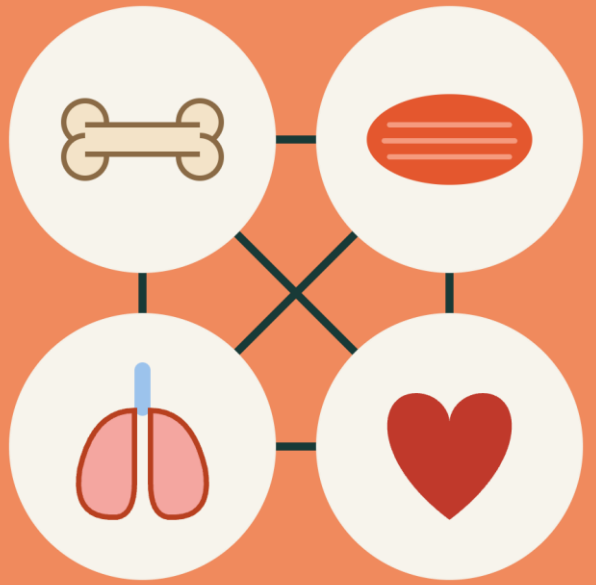
Hungry caterpillars

Caterpillars ate most of Kamatis’s leaves.

Effect: Less sugar is made, so the plant grows slowly and bears few tomatoes.

PART 2 · BODIES ARE TEAMS TOO

Your body is a team. When one player gets hurt, the whole team feels it.



Let’s follow Paolo and Bea to see how.

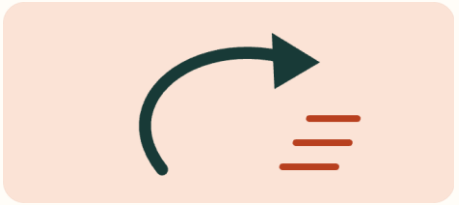
PART 2 · CASE 1: A TORN ACL

Paolo’s big game



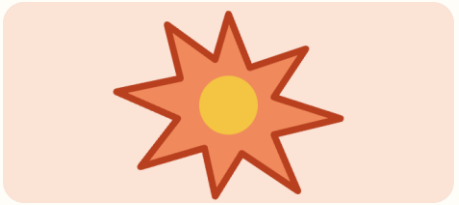
1

It’s the intramurals final. Paolo drives toward the basket.



2

He leaps, lands, and suddenly twists to dodge a defender.



3

“POP!” His knee buckles, and he crashes to the floor.



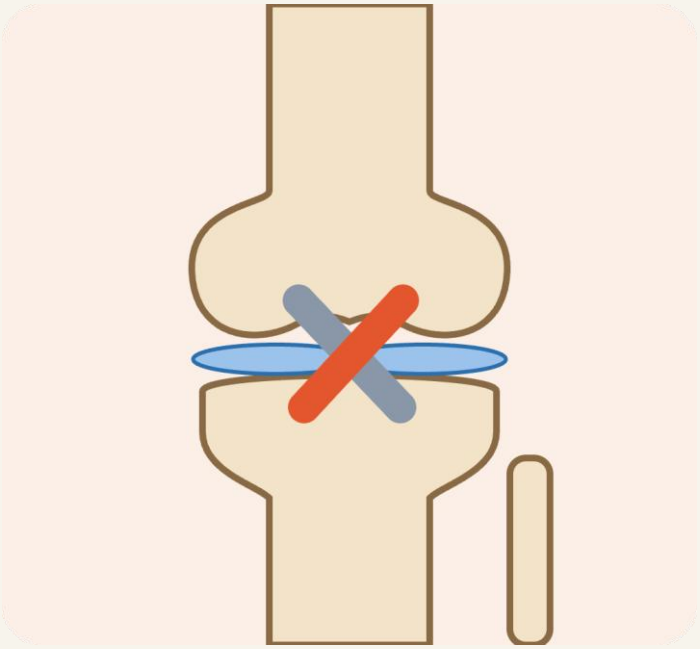
4

Within hours, his knee swells up. It feels loose and wobbly.

What just happened inside Paolo’s knee?

PART 2 · INSIDE THE KNEE

Meet the ACL



Bones
ACL
PCL (its partner)
Meniscus (cushion)

What is it? The ACL (anterior cruciate ligament) is a tough band inside the knee. It connects the **femur** (thighbone) to the **tibia** (shinbone).

Its job: It stops the shinbone from sliding forward and twisting too far, so the knee stays steady when you run, jump, and turn.

Teamwork: Muscles pull on bones through **tendons**. **Ligaments** hold bones together at the joint, so each pull becomes smooth, safe movement.

PART 2 · THE CHAIN REACTION

One torn band, many problems

1 · The damaged part	Paolo’s ACL is torn, so his knee loses its main “seatbelt.”
2 · Skeletal system	The knee joint becomes loose. The shinbone slides and twists, which can tear the meniscus and wear down cartilage.
3 · Muscular system	Paolo avoids using his leg, so his thigh muscles (quadriceps) shrink and weaken. Other muscles must overwork.
4 · Whole organism	Paolo limps, uses crutches, and can’t play for months. Even his mood is affected.

PART 2 · HEALING

Paolo's road to recovery

1

Right away

First aid: **R**est, **I**ce, **C**ompression, **E**levation (RICE)

2

First week

A doctor examines the knee. An MRI scan shows the torn ACL.

3

Weeks later

Surgery builds a new ACL from a graft, often a piece of tendon.

4

For months

Physical therapy rebuilds muscle strength and balance.

5

About 9–12 months

Paolo returns to basketball once his knee is strong and stable.

Protect your knees

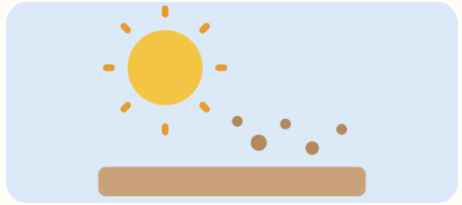
Warm up before every game.

Build strong thigh and hip muscles.

Land softly with bent knees.

PART 3 · CASE 2: AN ASTHMA ATTACK

Bea's PE class



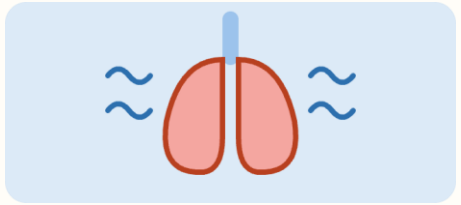
1

It's a hot, dusty afternoon. Bea jogs around the field in PE.



2

Smoke drifts over from someone burning dry leaves nearby.



3

Bea starts coughing and wheezing. Her chest feels tight.



4

She can't finish a sentence without stopping to gasp for air.

What is happening inside Bea's airways?

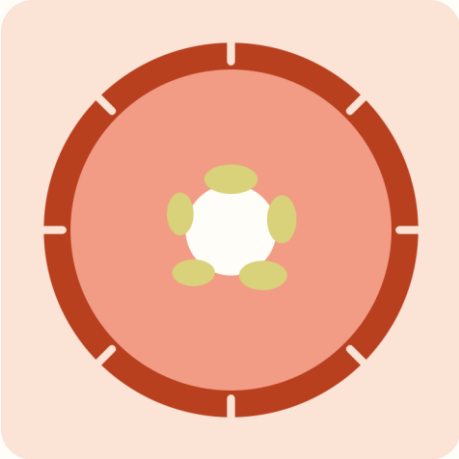
PART 3 · INSIDE THE AIRWAYS

The squeezed straw



Normal airway

Wide open. Air flows easily.



During an attack

Narrow. Air barely gets through.

1. **Muscles tighten** around the airways and squeeze them.
2. **The lining swells** and becomes inflamed.
3. **Extra mucus** clogs the small opening that's left.

It's like breathing through a coffee stirrer instead of a straw.

PART 3 · THE CHAIN REACTION

Narrow airways, a tired body

1 · The trigger	Smoke and dust irritate Bea’s sensitive airways, and they narrow.
2 · Respiratory system	Less air reaches the alveoli (air sacs), so less oxygen enters her blood. Stale air full of carbon dioxide gets trapped.
3 · Circulatory system	Her blood carries less oxygen. Her heart beats faster, trying to deliver enough oxygen to every cell.
4 · Cells and whole body	With less oxygen, her cells make less ATP. Bea feels weak and dizzy. In severe cases, her lips turn bluish.

PART 3 · GETTING HELP

What helps Bea breathe again

Reliever inhaler

Quick relief. The medicine relaxes the tight airway muscles within minutes.

Controller medicine

Used every day, as the doctor prescribes. It calms the swelling over time.

Avoid triggers

Stay away from smoke and dust. Wear a mask on smoky days.

How to help a classmate: Stay calm, help them sit upright, help them use their inhaler, and tell a teacher right away.

Call 911 if: their lips or fingertips turn blue, they can’t speak in full sentences, or the inhaler doesn’t help.

PART 4 · THE BIG PICTURE

The domino effect



A part is damaged

A torn ACL, swollen airways, or a ring of missing bark



Its system struggles

The joint wobbles, the airways narrow, or sugar stops flowing



Partner systems strain

Muscles weaken, the heart races, or the roots starve



The organism suffers

Paolo can’t play, Bea can’t breathe well, and the tree slowly dies

Systems depend on each other, so damage in one part spreads.

PART 4 · COMPARE

Two cases, side by side

Feature	Torn ACL (Paolo)	Asthma attack (Bea)
Damaged part	A ligament inside the knee	The airways (bronchi and bronchioles)
Main system	Skeletal	Respiratory
Partner system	Muscular	Circulatory
What goes wrong	The knee joint becomes loose and unstable.	Narrow airways let in less air and oxygen.
Effect on the body	Limping, weak thigh muscles, no sports for months	Wheezing, a racing heart, weakness, and dizziness
How it gets better	RICE, surgery, and months of physical therapy	A reliever inhaler works in minutes; daily care prevents attacks.

PART 4 · CHECK YOUR UNDERSTANDING

Quick check: try it!

1. Which plant tissue carries water from the roots up to the leaves?

2. Why does a tree slowly die when a ring of bark is removed all around its trunk?

3. Name one effect of Paolo’s torn ACL on his skeletal system and one on his muscular system.

4. During Bea’s asthma attack, why does her heart beat faster?

Bonus: How is transpiration like sipping juice through a straw?

EXIT TASK

Be a body detective!

For each case, name the **damaged part**, the **systems involved**, and the **effect on the whole organism**.

CASE A

The chewed roots

Beetle grubs ate the roots of a young mango seedling. Now its leaves are wilting.

CASE B

The twisted ankle

Ana sprained her ankle in volleyball. Its ligaments are stretched and partly torn.

CASE C

The tired Lolo

Lolo’s weak heart pumps poorly. He gets out of breath climbing the stairs.

Every part matters. Take care of your team!